# CLINICAL PARAMETER ANALYSIS TO SHOW

# CORRELATION WITH DIABETES OUTCOME

# Business understanding

Objective

- This project aims to create a machine learning model which detects clinical parameters that are most significant in predicting/detecting diabetes.

# Key questions

- 1. Which parameter is the highest predictor of diabetes?
- 2. Which parameter is the lowest predictor of diabetes?
- 3. Which 2 parameters combined are likely to result in diabetes?
- 4. Which parameter is strongly correlated with diabetes outcome?

# Key stakeholders

- 1.Physicians
- Responsible for screening, diagnosing and managing patients.
- 2.Nutritionists
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- Responsible for providing dietary advise to patients especially those having obesity
- 3.Public health professionals
- Responsible for creating awareness on non communicable diseases like diabetes.

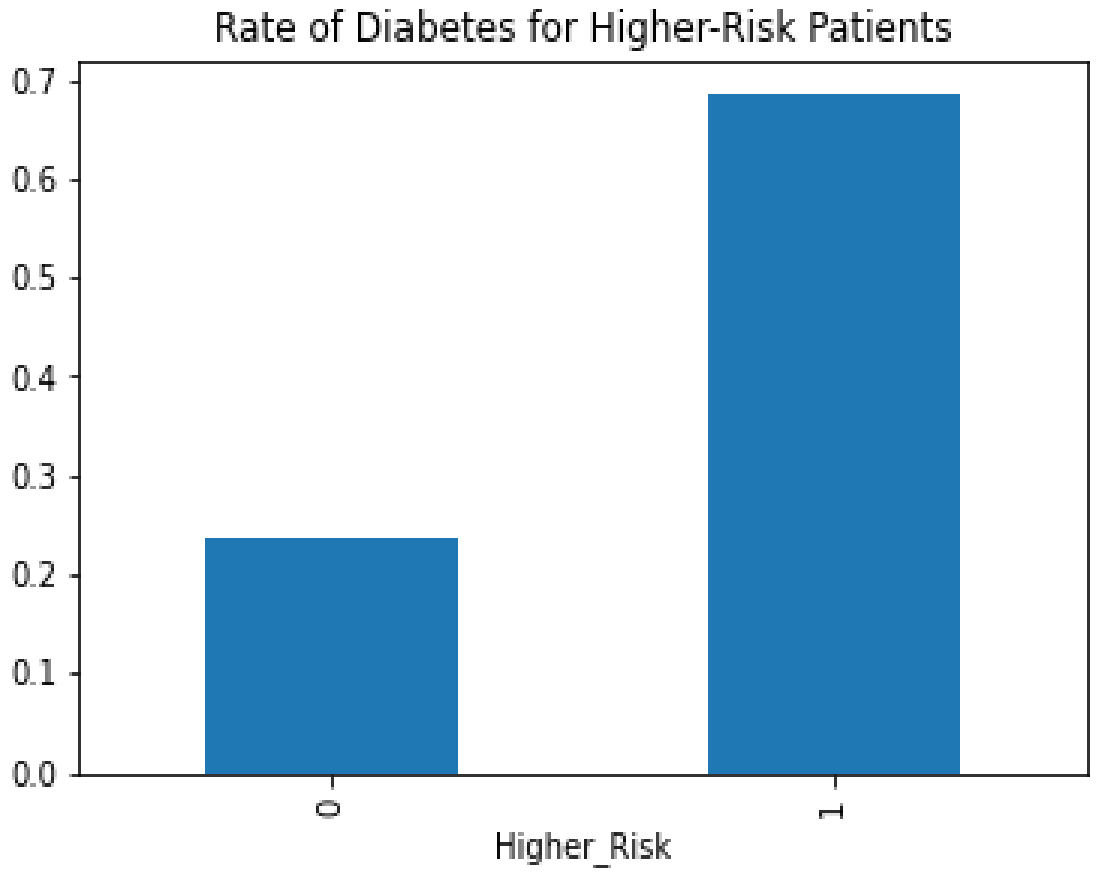
# Data Understanding

- The dataset used was obtained from Kaggle.

- It contains data of female patients aged atleast 21yrs and of Pima Indian origin.

- It contains 768 rows and 9 columns.

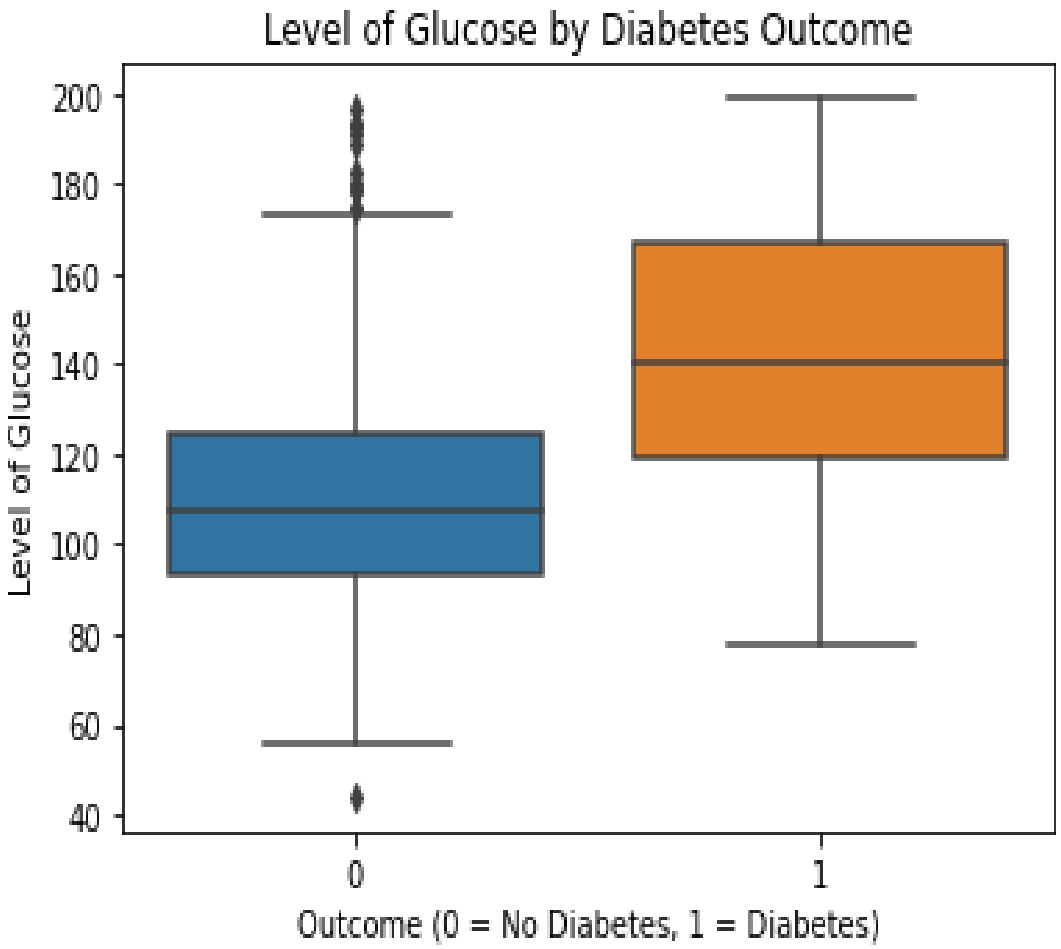
# Higher Rate of Diabetes for Higher-Risk Patients Risk



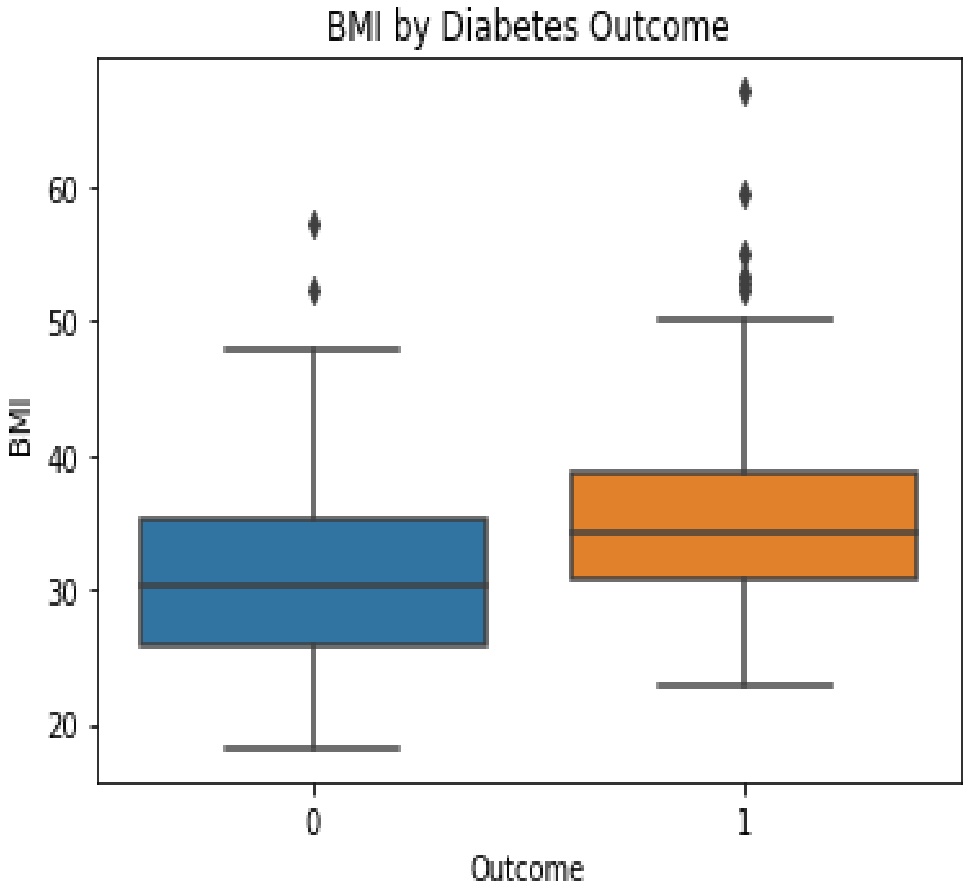

Patients with high levels of glucose and BMI>25 are at a higher risk of having diabetes as compared to having either parameters singly.

# Level of Glucose by Diabetes Outcome

Glucose levels is a key diagnostic feature for diabetes as most diabetic patients usually have high glucose levels



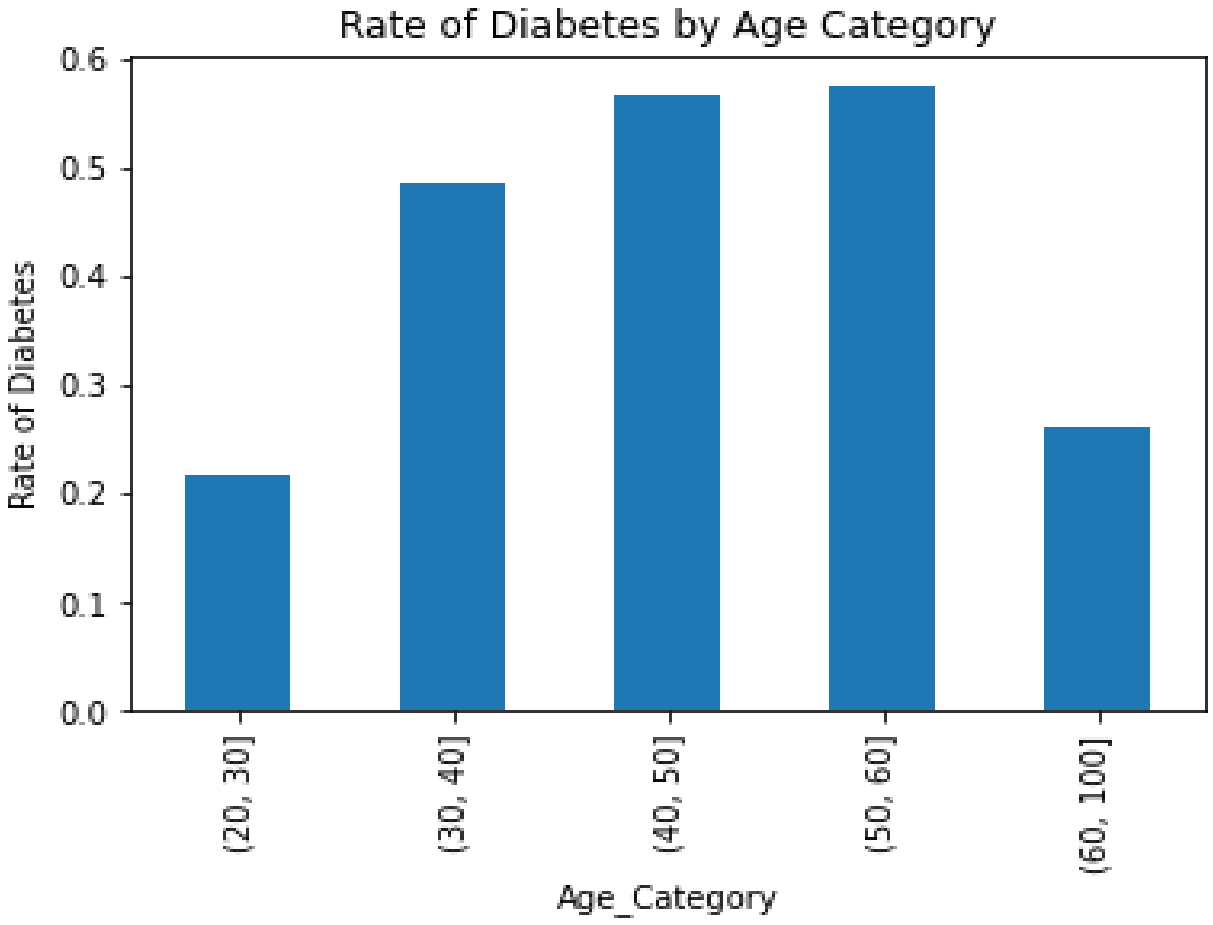
# BMI by Diabetes Outcome



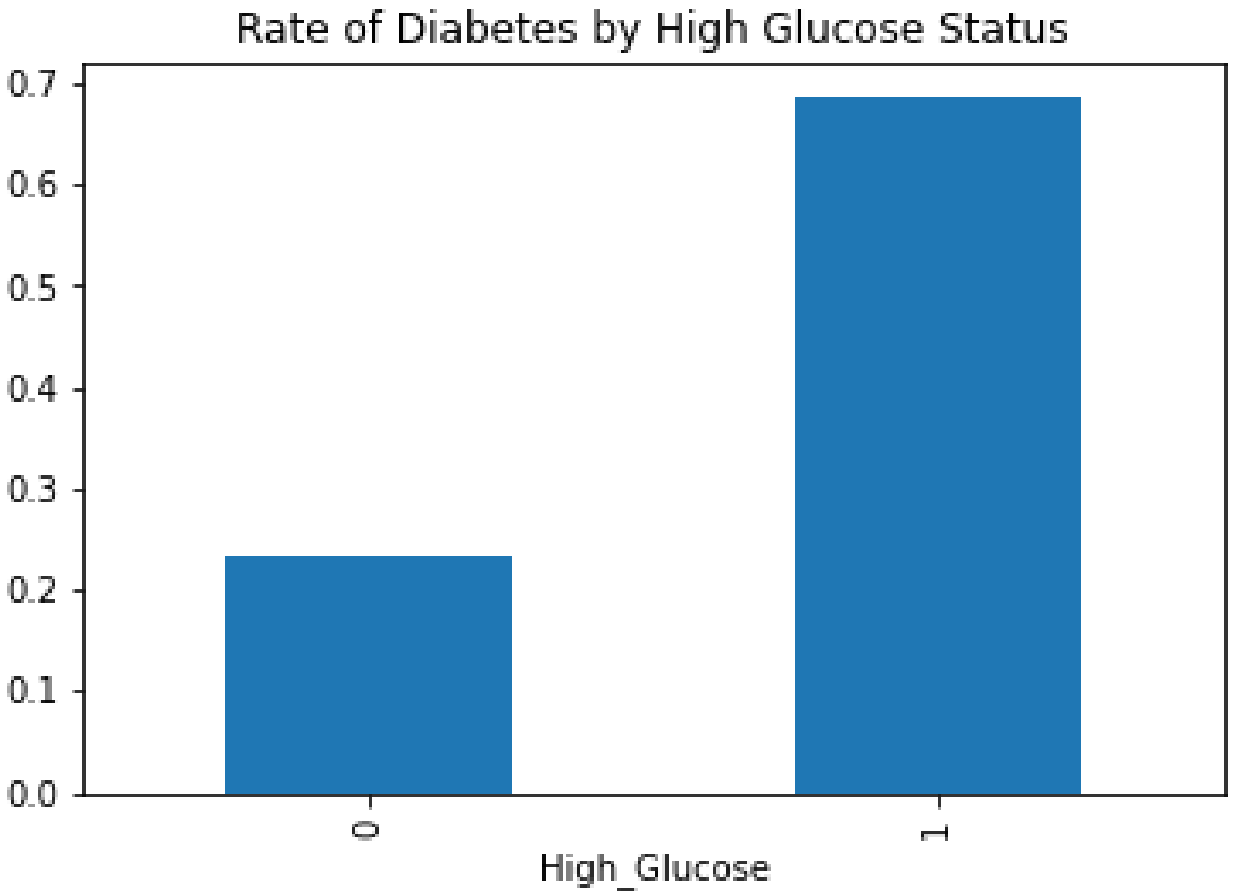

Obesity is a  major risk factor for diabetes as observed in the box plot.

# Rate of Diabetes by Age Category

People aged 40-60 yrs are at a higher risk of developing diabetes



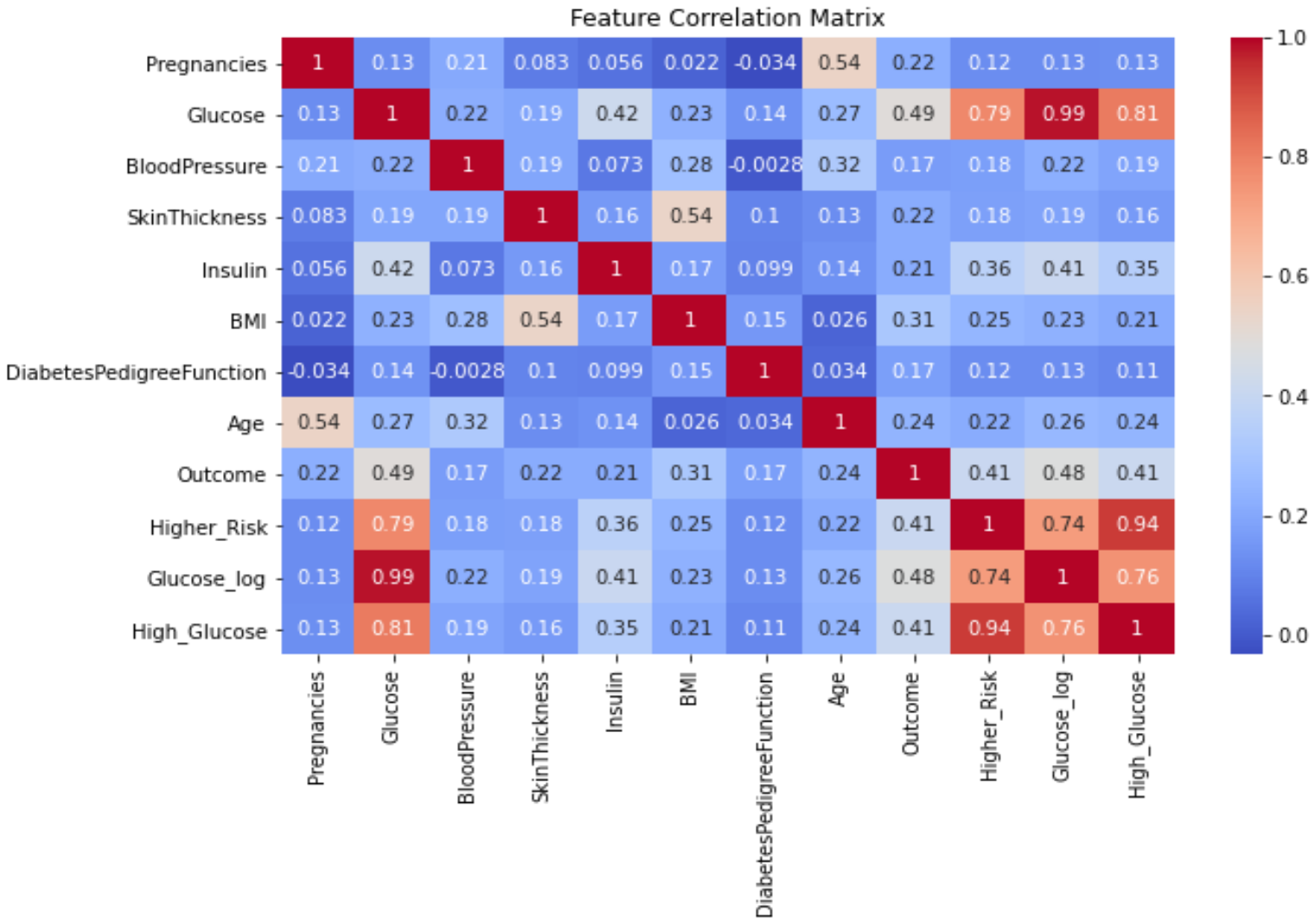
# Rate of Diabetes by High Glucose Status




Patients with high levels of glucose have high diabetes rates compared to others as observed in the bar chart

# Feature Correlation Matrix

Glucose is strongly correlated with diabetes outcome



# Findings

- Glucose is strongly correlated with diabetes outcome
- Glucose is the strongest predictor of diabetes
- Skin thickness is the least predictor of diabetes.
- BMI and Glucose combined have a high prevalence of diabetes outcome

# Recommendations

- Patients aged 40-60 should have their glucose levels checked routinely during hospital visits.

- Patients with a high BMI>25 should be put on weight reduction programs to prevent development of diabetes.